

Module 4: Troubleshooting and Helpdesk

Section 1 : Multiple Choice

1) What is the first step in the troubleshooting process ?

Ans : b) Identifying the problem

2) Which of the following tools is commonly used to diagnose hardware issues by testing electrical connections ?

Ans : c) Multimeter

3) Which Windows utility can be used to view system logs, monitor performance, and diagnose hardware and software issues ?

Ans : c) Event Viewer

Section 2 : True or False

4) Safe Mode is a diagnostic mode in Windows that loads only essential system services and drivers.

Ans : True

5) A system restore point is a snapshot of system files, registry, and settings that can restore the system to a previous state.

Ans : True

- 6) Ping is a command-line utility used to test network connectivity by sending ICMP echo requests and waiting for replies.

Ans : True

Section 3 : Short Answer

- 7) Describe the steps involved in troubleshooting a computer that fails to boot into the operating system.

Ans :

1) Check Power and Connections

- Make sure the computer is powered on.
- Check power cables, monitor cables, and connections inside the case.

2) Listen for Beep Codes or Error Messages

- Beep sounds or messages on the screen can indicate faulty RAM, CPU, or other hardware problems.

3) Check Boot Devices

- Enter BIOS/UEFI and make sure the correct drive (HDD/SSD) is selected as the boot device.

4) Disconnect External Devices

- Remove USB drives, external hard drives, and unnecessary peripherals that may disrupt booting.

5) Test RAM and Storage

- Reseat RAM modules or try one stick at a time.
- Check if the hard drive/SSD is detected in BIOS.

6) Boot into Safe Mode or Recovery Mode

- If the OS starts but fails, use Safe Mode or Windows Recovery to repair startup issues.

7) Run Startup Repair

- Use the operating system's built-in repair tool to fix damaged system files.

8) Check for Corrupted System Files

- Use tools like **SFC** or **CHKDSK** to repair errors.

9) Restore or Reinstall the OS

- As a last resort, perform a system restore or reinstall the operating system.

Section 4 : Practical Application

8) Demonstrate how to troubleshoot network connectivity issues on a Windows computer using the ipconfig command.

Ans :

1. Open Command Prompt

- Press **Windows + R**
- Type **cmd**
- Press **Enter**

2. Check Your IP Address

- Type the command : **ipconfig**
- This shows :
 - IP address
 - Subnet mask
 - Default gateway

3. Release the Old IP Address

- If the network is not working, release the current IP by typing : **ipconfig /release**

4. Renew the IP Address

- Now request a new IP from the router : `ipconfig /renew`
- This often fixes :
 - No internet
 - Limited connectivity
 - “Unidentified network”

5. Flush the DNS Cache

- If websites are not loading properly, type : `ipconfig /flushdns`
- This clears old DNS entries and can fix name-resolution problems.

6. Verify Network Configuration

- Type : `ipconfig /all`
- Check :
 - Whether the computer received an IP address
 - The correct DNS server
 - The correct gateway address

Section 5 : Essay

9) Discuss the importance of effective communication skills in a helpdesk or technical support role.

Ans :

- Effective communication skills are very important for anyone working in a helpdesk or technical support role. These professionals help users solve computer or technical problems, so they must be able to explain solutions clearly and politely.

1. Helps Understand the Problem

- Good communication allows support staff to ask the right questions. This helps them understand the user's issue quickly and accurately.

2. Reduces Confusion

- Many users are not technical. Clear communication helps avoid misunderstandings and ensures that instructions are easy to follow.

3. Builds Trust and Professionalism

- When support staff speak politely, patiently, and confidently, users feel comfortable and trust the support team. This creates a positive experience.

4. Improves Problem-Solving

- Communication skills help technicians give step-by-step instructions. Good listening also helps identify the real cause of the problem.

5. Increases Customer Satisfaction

- A friendly and helpful attitude makes users feel valued. Even if the issue takes time to solve, good communication keeps the user calm and satisfied.

6. Helps in Team Collaboration

- Technical support staff often work in teams. Effective communication helps share information, report issues, and work together efficiently.